



OPEN Cardiomyopathy and thrombogenesis in cats through left atrial morphological and fluid dynamics analysis

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The relationship between atrial fibrillation (AF), blood flow disturbances and thrombus formation is well-established in humans. Thrombi primarily form in the left atrial appendage (LAA), but the specific role of LAA morphology remains unclear. Felines, despite rarely experiencing AF, present a valuable model for studying human heart diseases. Cats exhibit a high incidence of cardiomyopathies, often accompanied by significant left atrial dilation and thrombus formation. This unique aspect prompts research focusing on LAA morphology and function in relation to thrombus development. Our study, using advanced image processing and fluid simulations on 24 cat hearts ranging from normal to cardiomyopathy subjects, aims to find distinctions in left atrium (LA) and LAA morphologies, blood flow patterns and their relation to thrombus formation. Significantly different variables in feline heart groups included: heart weight, LAA volume, LA volume, ostium area, ostium area-to-heart weight ratio, centreline length and velocity at the LAA ostium. Fluid simulations revealed lower LAA blood flow velocities and increased thrombotic risk in cases with thrombi. Our study enhances the understanding of LAA structure and function, offering insights into thrombus formation mechanisms with potential implications for better managing thrombosis risk in patients.

Cardiogenic thromboembolic events represent a significant health burden, leading to significant morbidity and mortality¹. A common trigger for thrombus formation within the left atrium (LA) is atrial fibrillation (AF), a cardiac arrhythmia characterized by erratic and ineffective LA contractions. The abnormal LA wall motion promotes blood stasis, particularly within the left atrial appendage (LAA), commonly originating thrombi. However, the precise impact of LAA morphology and function on the thrombogenesis process remains incompletely understood, with many aspects yet to be fully elucidated.

Cats, with their high incidence of spontaneously occurring heart diseases, have emerged as a valuable animal model for comprehending human cardiovascular conditions². In fact, cats exhibit a higher propensity for thrombus formation compared to other species such as dogs and humans^{3–7}. Among feline heart diseases, cardiomyopathies are the most prevalent, often correlated with progressive LA dilation, blood stasis within the LAA and subsequent thrombus formation, culminating in arterial thromboembolic events. Unlike humans, cats rarely develop AF, making them an ideal model for investigating the influence of LAA morphology and function on thrombus formation, independent of arrhythmias.

Previous studies have established that left atrial size, as assessed through echocardiography, is an independent predictor of arterial thromboembolism in cats, with a 48% incidence rate observed in those with hypertrophic cardiomyopathy (HCM)⁸. Feline cardiac function is assessed with echocardiographic imaging through 2D morphological measurements. Typically, a heart is classified as normal when the maximum diameter of the left

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atrium (LAD) is around 16 mm, while a LAD exceeding 25 mm is associated with an increased thromboembolic risk⁹. Notably, significant differences in left atrial volume (LAV) have been documented between healthy cats and those suffering from congestive heart failure (CHF): LAV values ranging from 0.53 mL to 1.77 mL in healthy cats and from 2.72 mL to 3.87 mL in those with CHF^{10,11}.

To gain a better understanding of the relationship between feline LAA morphology, haemodynamic characteristics and the development of thrombi, traditional measurements alone prove insufficient. While standard metrics provide valuable insights, additional parameters are required. In the context of human studies, advanced metrics such as ostium dimensions (the ostium being the interface between the LA and LAA) and LAA centreline analyses have been employed to deepen our understanding of LA and LAA morphology in relation to thrombus formation¹². Furthermore, computational fluid simulations (CFD) have provided invaluable insights into the 4D nature of blood flow patterns within the LA, offering in-silico metrics for assessing thrombogenic risk^{13–18}. Thus far, these advanced techniques have not been used in the context of evaluating the cardiac function of feline hearts. We therefore seek to advance our understanding of feline left atrial structures through an extensive morphological analysis and mathematical modelling of micro-computed tomography (micro-CT) data extracted from 24 feline hearts. Our aim was to elucidate distinctions between healthy feline hearts (n = 8) and pathological cases (n = 16), with and without thrombi, including conditions such as CHF and asymptomatic hypertrophic cardiomyopathy (HCM).

Results
Morphological analysis

Table 1 provides a comprehensive overview of our cohort, stratified into two primary categories: normal and cardiomyopathic hearts. Within the cardiomyopathy cohort, the analysis is twofold. Firstly, we separated these cases into those afflicted by thrombus formation and those without; secondly, we classified them as either symptomatic or asymptomatic individuals.

Normal vs cardiomyopathy

Our investigation revealed notable differences across multiple parameters when comparing normal and cardiomyopathic cases. In particular, significant distinctions were evident in heart weight, LAA and LA volumes, ostium area, centreline length and ostium velocity. One salient finding was the enlarged ostium area observed in cardiomyopathy cases in comparison to normal cases. Importantly, this expansion of the ostium area was linked to lower velocities within the cardiomyopathy group as opposed to within the normal group.

	Normal	Cardiomyopathy	p-value	Cardiomyopathy					
				Thrombus	Non-thrombus	p-value	Asymptomatic	Symptomatic	p-value
Gender [n] *	Female (2)	Female (5)	1.00	Female (3)	Female (2)	0.58	Female (1)	Female (4)	1.00
	Male (3)	Male (10)		Male (3)	Male (7)		Male (3)	Male (7)	
Neutered [n] *	Yes (1)	Yes (12)	0.13	Yes (4)	Yes (8)	0.53	Yes (3)	Yes (9)	1.00
	No (2)	No (3)		No (2)	No (1)		No (1)	No (2)	
Age *	4.63 ± 2.84	7.93±3.75	0.46	8.33 ± 3.08	7.67±4.30	0.73	8.75 ± 6.60	7.64 ± 2.50	0.76
Heart weight [g]	14.36 ± 2.80	22.41 ± 6.62	0.001	24.07 ± 8.50	21.42 ± 5.48	0.52	20.16 ± 6.47	23.44 ± 6.73	0.38
LAAv	0.28 ± 0.09	1.07 ± 0.85	p<0.001	1.76 ± 1.07	0.67 ± 0.30	0.02	0.57 ± 0.31	1.31 ± 0.93	0.05
LAv	1.28 ± 0.49	3.78 ± 1.99	p<0.001	5.23 ± 2.08	2.92 ± 1.39	0.04	2.31 ± 1.34	4.45 ± 1.90	0.03
LAAv/LAv [%]	24.45 ± 9.77	27.59 ± 14.79	0.70	35.02 ± 23.15	23.14 ± 2.79	0.49	24.61 ± 3.16	28.95 ± 17.82	0.66
OA	44.59 ± 2.76	117.60 ± 55.08	p<0.001	157.92 ± 65.66	93.41 ± 30.45	0.04	92.66 ± 40.67	128.94 ± 58.62	0.15
OA/LAv	38.16 ± 11.04	33.75 ± 10.11	0.89	31.56 ± 11.75	35.06 ± 9.42	0.55	42.34 ± 6.76	29.84 ± 9.02	0.01
OA/LAAv	171.92 ± 53.39	134.54 ± 45.52	0.20	105.50 ± 38.32	151.97 ± 41.69	0.04	175.38 ± 44.94	115.98 ± 32.98	0.04
OA/HW	3.19 ± 0.59	5.60± 3.03	0.002	7.06 ± 3.31	4.72 ± 2.62	0.18	5.19 ± 3.71	5.78 ± 2.85	0.51
CL [mm]	7.79 ± 2.18	14.46 ± 5.60	0.01	19.42 ± 4.31	11.49 ± 3.97	0.004	9.41 ± 2.03	16.76 ± 5.18	0.001
τ	0.84 ± 0.07	0.73 ± 0.24	0.31	0.60 ± 0.26	0.81 ± 0.20	0.02	0.89 ± 0.06	0.65 ± 0.26	0.01
Ostium velocity [m/s]	0.12 ± 0.03	0.07 ± 0.06	0.01	0.03 ± 0.03	0.09 ± 0.06	0.04	0.12 ± 0.07	0.04 ± 0.03	0.02
ECAP	5.61 ± 2.43	6.04 ± 4.75	0.72	5.98 ± 3.93	6.08 ± 5.39	0.58	4.78 ± 3.78	6.62 ± 5.20	0.56

Table 1. Cohort characteristics and left atrial appendage (LAA) morphological indices of normal (n = 5) and cardiomyopathy feline hearts (n = 16). The latter were analysed as: asymptomatic (n = 5) vs. symptomatic cats (n = 11), and cats both with (n = 6) and without (n = 10) thromboembolism. LAv/LAAv (mL): left atrial and appendage volumes; OA (mm²): ostium area; CL (mm) and τ: LAA centreline length and tortuosity, respectively; Age (years). *Data was not available for all cases. Haemodynamic indices in the left atria for normal (n = 5) versus cardiomyopathy cats (n = 16), asymptomatic (n = 5) versus congestive heart failure (CHF) and thrombus (n = 11), as well as for non-thrombus (n = 10) versus thrombus (n = 6). Results are presented as mean standard deviation. Statistically significant values are in bold.

Thrombus vs non-thrombus

Upon closer examination of the cardiomyopathy cases, as they were stratified into thrombus and non-thrombus groups, notable discrepancies were present in LAA and LA volumes, ostium area, centreline length and ostium velocity. Within the thrombus-afflicted cases, a significant increase in ostium area was evident in comparison to those without thrombus. This finding aligns with previous human studies, where larger ostium dimensions have been associated with reduced velocities within the LAA, potentially contributing to the phenomenon of blood stasis¹⁹.

Symptomatic vs asymptomatic

In addition to the differences previously discussed, symptomatic cases in the cardiomyopathy group exhibited distinctive features. These individuals displayed longer and less convoluted LAA centrelines, alongside elevated LA and LAA volumes and ostium areas. Moreover, when we normalized the ostium area with LA and LAA volumes, symptomatic cases exhibited lower values, possibly preventing a complete LAA washing and inducing blood stasis. Regrettably, due to the constraints of a limited sample size, our study did not yield statistically significant variations in clinical and morphological characteristics between CHF cases with and without thrombus formation. Only two cases fell within the former category, and a mere five cases within the latter, rendering it challenging to draw meaningful conclusions in this subset.

Haemodynamic study

Blood flow velocities

Figure 1 illustrates the average blood flow velocity profiles averaged across the LAA ostium for each feline heart category during the second cardiac cycle, obtained from computational fluid simulations. Furthermore, Fig. S4 in the Supplementary Material shows the individual velocity profiles for each individual case. We observed some expected trends: elevated velocities for normal cases, reduced values for thrombus cases and a middle-ground scenario for non-thrombus CHF cats. By contrast, among thrombus CHF individuals, the only two cases available in the studied cohort presented significant variations, hampering any meaningful conclusion in this category.

Table 1 displays the blood flow velocities at the LAA ostium, averaged across all available cases. Our analysis revealed statistically significant differences among the groups. Specifically, higher velocities were detected in the normal cases when compared to the cardiomyopathy group. Additionally, the thrombus cases exhibited noticeably lower velocities in comparison to the non-thrombus cases, a correlation that aligns with the elevated risk of thrombus formation. A similar pattern emerged when comparing the asymptomatic group to both CHF and thrombus cases. In these instances, lower velocities were observed for the CHF and thrombus-afflicted cats, indicating a potential correlation with the asymptomatic group's distinct haemodynamic profile.

Flow metrics at the left atrial wall

Figure 2 displays the spatial distribution of the endothelial cell activation potential (ECAP) index, a computational metric derived from fluid simulations that identifies regions characterized by complex flows (high Oscillatory Shear Index, OSI) and reduced velocities (low Time Averaged Wall Shear Stress, TAWSS), offering insight into thrombotic risk (red-green regions in the figure correspond to high ECAP values, indicating an elevated susceptibility to thrombus formation). Figure S2 and S3 in the Supplementary Material display the corresponding OSI and TAWSS maps, while Fig. S5 shows the streamlines of simulated blood flow patterns for a representative case of each type of heart disease, at two different times of the cardiac cycle (end-systole and end-diastole). In our study, owing to the abundant trabeculations within the LAA and the detailed resolution of the micro-CT

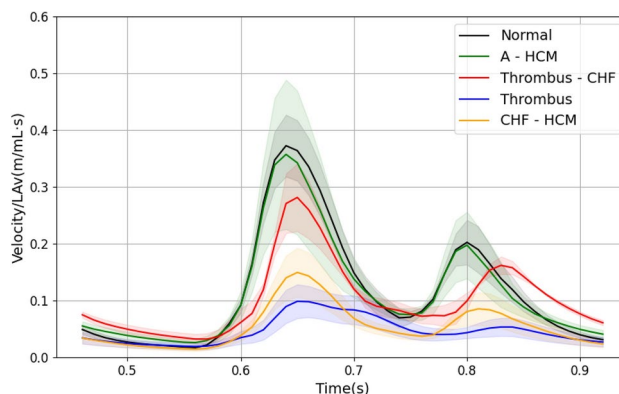


Fig. 1. Blood flow velocity profiles at the left atrial appendage (LAA) ostium (normalized by left atrial volume) from computational fluid simulations during the second cardiac cycle, for each category of feline hearts (solid line and shaded areas represent the average and the standard deviation of all cases in each subgroup, respectively). The studied feline heart categories are the following: normal, asymptomatic hypertrophic cardiomyopathy (A-HCM), thrombus with and without congenital heart failure with thrombus (Thrombus-CHF and Thrombus, respectively) and congenital heart failure with hypertrophic cardiomyopathy (CHF-HCM).

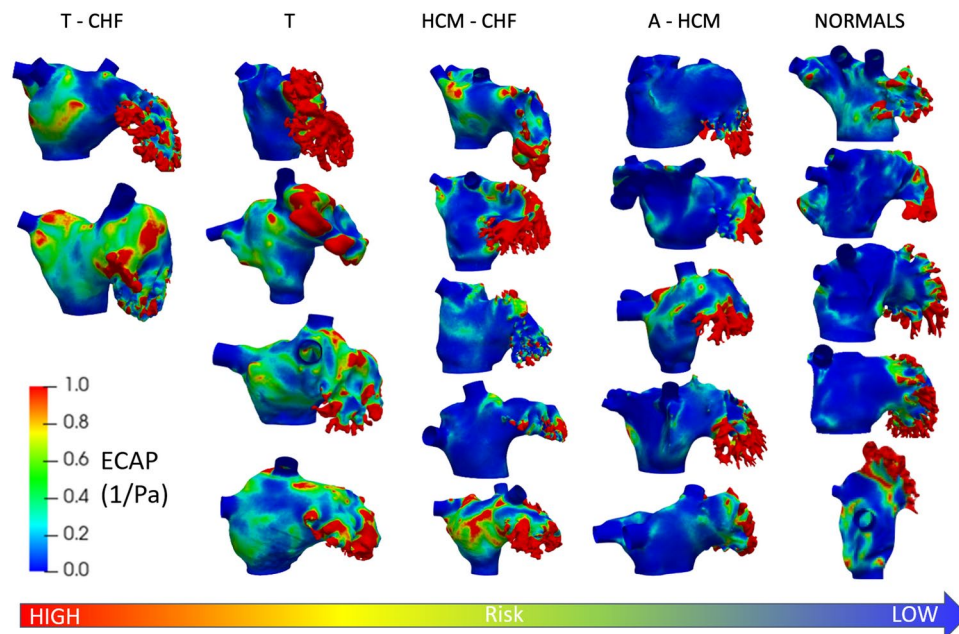


Fig. 2. Endothelial cell activation potential (ECAP) maps from the in-silico fluid simulations for each feline heart category. Red and green colours represent areas with high ECAP values (lower blood flow velocities and more complex flow), thus more prone to thrombus formation. CHF: congestive heart failure; A-HCM: asymptomatic hypertrophic cardiomyopathy; T: thrombus.

images, the LAA tip, situated farthest from the ostium, and its adjacent lobes, consistently appear in red on the ECAP maps. This red shading represents regions where there is a higher probability of finding a thrombus²⁰. Beyond the LAA tip, a discernible ECAP gradient becomes evident in most cases, starting from the lower thrombotic risk categories on the right side of Fig. 2 (uniform blue ECAP maps, indicative of high velocities, i.e., high TAWSS values, and laminar flow, i.e., low OSI values), corresponding to normal and asymptomatic cases. At the opposite end of the disease spectrum, feline hearts at higher risk of thrombus formation provide more heterogeneous ECAP maps in shades of red and green. Distinguishing between asymptomatic and symptomatic HCM cases is challenging, as with previous metrics. However, symptomatic HCM feline hearts display higher ECAP values around the LAA ostium, corroborated by Table 1, where asymptomatic cases exhibit higher blood flow velocities at the ostium. Additionally, more complex LA cavity flow patterns can be observed in the streamline visualization (Fig. S5) of some pathological cases such as in thrombus and CHF cases compared to normals, as it is also reflected by higher values in OSI maps (Fig. S2).

Higher velocities are observed within the LA, whereas the presence of trabeculations in the LAA leads to lower velocities in this region. Consequently, high values of TAWSS are concentrated in areas with higher velocities, while regions with trabeculations exhibit reduced TAWSS. In thrombus cases, moreover, TAWSS values are notably lower in the LA, which, when combined with reduced velocities, creates a favourable environment for thrombus formation.

Shear rate

In each feline heart group, the average shear rate is displayed in Fig. 3, with lower values signifying heightened blood viscosity and an associated increased risk of coagulation and thrombus formation. The normal group demonstrated significantly higher average shear rates in both the LAA and the LA when compared to the cardiomyopathy cats. In addition, when we contrasted the shear rates between asymptomatic and thrombus cases, as well as between non-thrombus and thrombus categories, it became apparent that the asymptomatic and non-thrombus cases exhibited higher shear rates.

Discussion

Our study aimed to assess left atrial morphology and haemodynamics in control cats as well as in cats with CHF and thrombus, using advanced morphological analysis and computational modelling techniques. Until now, such techniques have primarily been employed in human studies. As far as we know, this is the first study in which advanced haemodynamic simulations have been utilized to analyze feline heart diseases.

Morphological analysis

As anticipated, we observed significant differences in morphological indices, including the volumes of the LA and LAA, which were notably larger in cats with cardiomyopathy compared to their normal counterparts^{3,11,21}. Our findings on left atrial volumes align with those reported by Duler et al¹¹. Of particular interest, our study revealed larger LA ostia in diseased feline hearts, according with some studies performed in humans¹² but

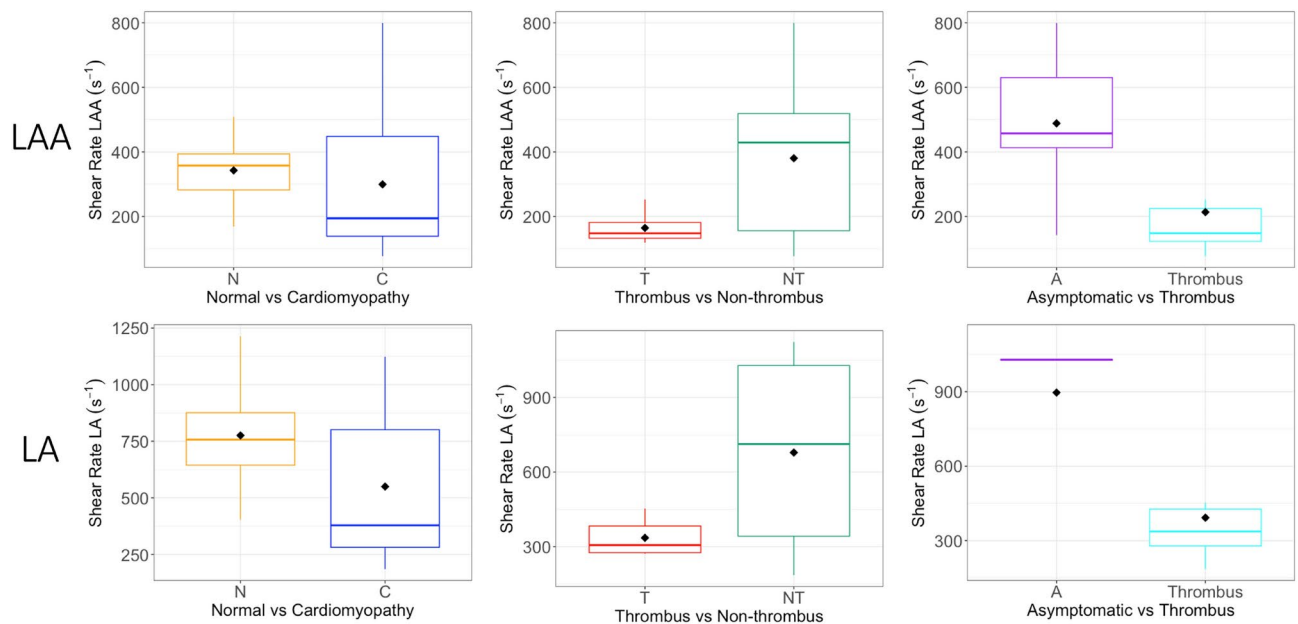


Fig. 3. Average shear rate values in the left atria (LA) and left atrial appendage (LAA) for each feline heart category, represented in boxes by groups (left to right): normal (N) vs cardiomyopathy (C) cases; thrombus (T) vs non-thrombus (NT) cases; asymptomatic (A) vs thrombus.

contradicting others such as Khurram et al.²² and Fang et al.²³. While larger-scale investigations are needed to establish definitive thresholds for atrial morphological indices, our results underscore the importance of these parameters in identifying cats at risk of developing cardiomyopathy.

Morphological variations between thrombus and non-thrombus groups were also observed. For instance, cats with thrombus had a larger LA and ostium. Furthermore, these cats exhibited longer LAA centrelines with lower tortuosity values, findings that align with results observed in human hearts²³. Similarly, cats with symptomatic cardiomyopathies (HCM and thrombus cases) presented a larger LA. Despite the significant differences (from p-values), it is worth noting that the variances in morphological parameters were substantial in relation to their magnitudes. This high variability complicates the establishment of precise threshold values. A previous study reported a decrease in LA ejection fraction in cats with HCM and CHF, accompanied by significant LA enlargement¹¹. However, such a decrease was not observed in cats with asymptomatic HCM. The causal relationship between decreased LA function and dilatation remains to be established. Our study indicates a correlation between larger LA size and diminished heart function in cats with cardiomyopathy. In cardiomyopathies, the LA does not experience significant hypertrophy; instead, it undergoes dilation. This dilation occurs in response to an increased afterload, as the left atrium must effectively empty into a stiff left ventricle. This stiffness results from a combination of hypertrophy and fibrosis.

Haemodynamics and risk of thrombus formation

The haemodynamic results provided valuable insights into feline heart phenotyping. Notably, the blood flow velocity in normal cases was significantly higher, approximately five times that of cats with cardiomyopathy (see Fig. 1). This difference can be attributed to blood flow stagnation, particularly in thrombus cases. We also observed lower blood flow velocities at the LAA ostium in the thrombus group compared to the non-thrombus group, as well as in symptomatic (CHF + thrombus) cases compared to asymptomatic cases. This pattern underscored a direct correlation between a larger ostium area (normalized by the LA volume) and reduced velocities in feline hearts. It is essential to note, however, that the data from the asymptomatic cases exhibited substantial variability involving significant overlap with various groups. This variability limits our ability to draw conclusive insights from this particular subgroup. Therefore, more advanced morphological characterization is still needed to better understand the complexities of blood flow dynamics and their implications for thrombus formation in feline hearts.

High thrombotic risk areas within the feline hearts were visually identified with fluid simulations, primarily based on regions with elevated ECAP values. These high ECAP regions are indicative of complex flow patterns and slower velocities, and are represented by the larger red-green areas in Fig. 2. Qualitatively, we observed that differences in ECAP distributions within the LAA were not readily discernible among different cases.

Higher ECAP values were consistently found at the tip of the LAA across all groups, probably because blood flow did not reach this portion of the LAA. It should be stated that these values might be influenced by the presence of trabeculae in the LAA. There is therefore a need for more robust methods to quantify the maps accurately. On the other hand, we observed that the highest ECAP values averaged over the entire LA, beyond the LAA, were found exclusively in pathological cases. Haemodynamic differences within the LAA were, moreover, distinctly observed among different groups using alternative in-silico indices such as the shear rate

(see Fig. 3); healthy cases displayed higher shear rate values, which has been associated with a reduced likelihood of thrombus formation²⁴.

Limitations and future work

The limited number of studies on non-human data in the existing literature restricts the ability to draw robust conclusions from our obtained results. Firstly, the large inter-species variability in the morphology of the LAA complicates direct comparisons of thrombogenesis and the establishment of normal reference values. For instance, in any in-silico modelling of haemodynamics and thrombus formation or when comparing across species, it is crucial to account for variations in blood properties. These variations include differences in the coagulation and fibrinolytic systems, as they significantly influence clot formation and resolution, consequently affecting the risk of thrombosis^{25,26}. Moreover, the different imaging and measurement modalities available for studying the LA in different species can introduce variability in the accuracy of results^{27,28}.

There are several limitations in our study in relation to animal handling, image acquisition, and available information for each cat. Handling *ex-vivo* specimens presents challenges, as it is difficult to preserve all relevant physiological conditions. The images were acquired post-mortem, and despite efforts to maintain anatomical accuracy, discrepancies may arise given the time between death and image acquisition. In parallel, the use of micro-CT imaging for model construction allowed for high-resolution visualization of fine anatomical details, such as the trabeculations of the LAA. However, manually segmenting the micro-CT scans introduced uncertainties in the heart geometry, due to the complexity of the anatomical structures involved. While we conducted thorough checks in the resulting segmentations with veterinary experts, and the obtained LA volumes were in line with values in the literature¹¹, some degree of uncertainty may still remain. Future research focused on feline hearts should prioritize larger, more balanced databases, ideally obtained through prospective studies, which employ standardized *ex-vivo* and *in-vivo* data acquisition and imaging protocols. The inclusion of regular CT scans and complementary functional imaging (i.e., echocardiography) would allow a more automatic image analysis workflow and subject-specific boundary conditions to enhance the reliability of the findings. At the same time, a quantitative analysis of the LAA trabeculae observed in micro-CT images will facilitate the study of its potential role as highlighting areas where clots can become lodged. Furthermore, studying hearts from other species will contribute to improving our understanding of the LAA and its relation with atrial fibrillation and thrombus formation (e.g., dog hearts more prone to arrhythmias but rarely having thrombus).

The fluid modelling pipeline was designed with the goal of simulating haemodynamics in a relatively large population (> 20 cases), thus modelling choices were driven by computational costs, at the expense of better accuracy of results. For instance, we only simulated three cardiac cycles, which already identified different trends among the feline groups, but longer simulations are needed to have more numerically stable results, as we recently demonstrated²⁹. The simplicity of the modelling pipeline, mainly in terms of mesh resolution, compared to the high anatomical complexity in some regions of the LA (e.g., LAA trabeculae) provides a plausible explanation for the simulation divergence of three normal cases. The main difference in this study compared to our previous simulations using human data is the availability of micro-CT images for *ex-vivo* feline hearts, which offer a higher resolution than *in-vivo* CT scans, allowing for the identification of trabeculae. However, this increased resolution also makes the segmentation and meshing process more challenging and tedious, being difficult to obtain good-quality geometrical domains for reasonable computational times of the simulations. Alternative schemes, such as those using immersed boundary techniques³⁰, should be explored to address these issues. Additionally, a variety of metrics, including a separate analysis in different phases of the cardiac cycle, are necessary to properly quantify haemodynamics results. The variability in ECAP values may limit its capacity to fully capture the complex haemodynamics of the LA and LAA. As such, combining multiple indices is essential to more accurately estimate blood flow dynamics. In our study, we have mainly focused on the ECAP, obtained from the TAWSS and the OSI, but additional indices such as residence times³¹, blood ageing³², particle analysis¹⁸, and flow component analysis³³ would certainly contribute to a better understanding of stasis in the analysed cohort. These indices would collectively provide a more comprehensive characterization of blood flow patterns. Furthermore, thresholds typically employed for haemodynamic indices in human studies to determine risk (e.g., 0.2 m/s in ostium velocities) require a reassessment for non-human data. Finally, as in human studies, further research is needed to produce more realistic LA wall motion dynamics when time-resolved CT data is not available.

Conclusions

Despite the inherent limitations of our study, we observed consistent trends in key morphological characteristics and haemodynamic indices between feline and human hearts. These trends included large LA/LAA and ostium volumes, as well as large LAA length and tortuosity in disease cases, being associated with reduced blood flow velocities and higher risk of thrombus formation^{13,34,35}. The estimation and integration of morphological and in-silico haemodynamic indices in feline hearts was achieved through the adaptation of advanced computational tools, traditionally applied in human research. As a result, different phenotypes of feline hearts in relation to thrombus formation risk were found. Further refinement and customization of these computational tools hold promise for accurately predicting thrombus risk in asymptomatic cats, representing a substantial advancement in veterinary medicine. These tools also offer valuable insights for enhancing our understanding of the human LA and related disorders such as atrial fibrillation, through the analysis of non-human data.

Methods

Sample description

The prospective observational post-mortem study received ethical approval from the Clinical Research Ethical Review Board of the Royal Veterinary College (RVC) (URN 2016-1638-3). The clinical procedures of euthanasia

and post-mortem examination were conducted in accordance with the Veterinary Surgeons Act as part of recognised veterinary practice. Prior to the procedures, owners provided informed consent for the use of heart tissue obtained for research purposes. The College's Clinical Research Ethical Review Board meticulously reviewed the consent form and information provided to the owners regarding the project, ensuring they were fully informed of the research involving tissues obtained post-mortem. This study was designed as a prospective observational postmortem study.

Ex-vivo isolated feline heart specimens were evaluated by micro-computed tomography (micro-CT) assessment at Great Ormond Street Hospital (GOSH), followed by conventional gross and histopathology examinations, performed by a board-certified veterinary pathologist at the RVC, which served as the gold standard reference. A total of 24 hearts from pet cats that had died or been euthanized for medical reasons, with owner-informed consent, were included in the study. The control group ($n = 8$) consisted of normal hearts (N) from cats undergoing routine necropsy by the RVC Pathology Service. However, fluid simulations of three normal hearts were discarded due to the divergence of the results. The HCM group ($n = 16$) consisted of hearts from clinical cases of HCM diagnosed by veterinary cardiologists. Within the cardiomyopathy group, there were 5 cats with CHF (no thrombus), 5 asymptomatic cats and 6 cats with thromboembolism (two of them being diagnosed with CHF, then being studied as a separate subgroup).

The hearts were promptly excised after natural death or euthanasia with IV pentobarbital sodium, and gently flushed with water to eliminate any clots. Subsequently, they were immersed in 10% formaldehyde solution at a tissue-to-volume ratio of 1:20 for over 48 hours, followed by weighting. They were then submerged in a 2.5% potassium tri-iodide (I₂KI) solution with a tissue-to-volume ratio of 1:30 at room temperature for over 48 hours. After iodination, the hearts underwent a thorough water rinse to remove any residual contrast medium, followed by gentle towel-drying.

As detailed in Simcock et al.³⁶, systematic procedures for micro-CT imaging and staining protocols were carried out in the feline hearts under investigation. This imaging process utilized a Nikon XTH225 ST and a Med-X micro-CT scanner (Nikon Metrology, UK). Geometric data were reconstructed employing the modified Feldkamp filtered back projection algorithm, implemented through proprietary software (CTPro3D; NikoMetrology), and further post-processed using VGStudio MAX (Volume Graphics GmbH, Heidelberg, Germany). Subsequently, each heart was comprehensively reconstructed from the micro-CT images into isotropic volumes, with voxel sizes ranging from 14 to 24 μm . This reconstruction process yielded approximately 900 two-dimensional slices for each case, respectively.

Data pre-processing

The segmentation of the entire heart from micro-CT scans was accomplished using a custom in-house script developed in Matlab R2018b (R2018b Academic license; <https://matlab.mathworks.com>). This automated procedure integrated Otsu thresholding techniques, 3D Gaussian smoothing (7 pixels per dimension) and morphological operations to minimize noise. Following this, the LA was manually delineated and 3D surface meshes were generated using the marching cube algorithm with the LA binary masks (see Fig. 4).

For fluid simulations, volumetric meshes were generated following a computational workflow inspired by Aguado et al.¹³. The mesh resolution ($\Delta x = 0.2 \text{ mm}$) was chosen considering computational times and based on prior human heart studies^{12,15,29,37}, which included detailed sensitivity analyses and mesh dependency tests following V&V40 guidelines. The resulting number of tetrahedral elements across the meshes ranged between 500K and 600K, depending on the size and complexity of the anatomical geometry.

Geometries underwent smoothing in MeshLab 2016.12 (<https://www.meshlab.net/>) employing a Taubin filter ($\lambda = 0.6$, $\mu = -0.53$) and iterating the process ten times. The pulmonary veins (PV) and the mitral valve (MV) were created with artificial cylinders using MeshMixer 3.5. (<https://www.meshmixer.com>). Finally, Gmsh (<https://www.gmsh.info>) was utilized to construct the volumetric meshes.

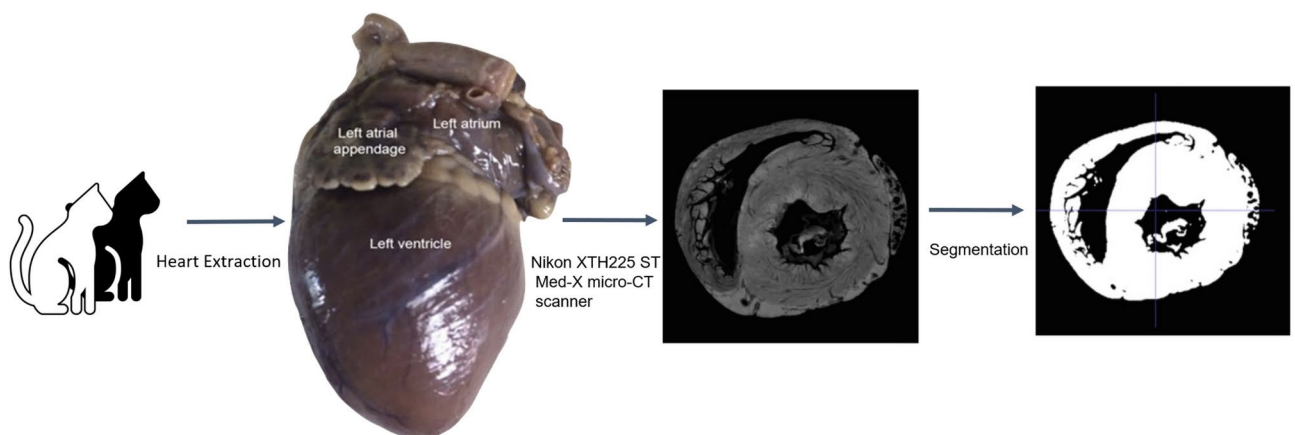


Fig. 4. Feline hearts were isolated *ex-vivo* and evaluated with micro-computed tomography (micro-CT) images, from which the left atria were segmented prior to creating 3D surface meshes.

Morphological indices

Various shape descriptors were computed using an in-house script to characterise the morphology of the LA and LAA. These descriptors included:

1. The volume of the LA and LAA.
2. The ratio between the volumes of LAA and LA (LAA/LA).
3. Manual selection of the ostium, following a protocol validated in human data³⁸, from which the following parameters were calculated: area of the ostium (OA) and the maximum and minimum diameter (Dmax, Dmin, respectively) which are known to be associated with the risk of thrombus formation in humans¹⁹, as illustrated in Fig. 5.

Additionally, the centreline (CL) of the LAA was defined as a line connecting the centre of the ostium to the farthest geodesic point within the LAA, as described by Genua et al.³⁹. Several descriptors were then derived from this centreline, including its total length and its tortuosity (τ), defined as the total length divided by the height of the LAA (L). Finally, several morphological descriptors were normalized by the volume of the LA and LAA, as well as heart weight, to take into account different heart sizes.

Computational fluid dynamics simulations

Computational fluid dynamic simulations (CFD) employing the finite-volume method (FVM) were conducted using the Ansys Fluent 2019 solver (Ansys Inc, USA; <http://www.ansys.com>). In the simulations, blood was modelled as an incompressible Newtonian fluid with a density of 1060 kg/m³ and a dynamic viscosity of 0.0035 Pa s. This modelling was executed through the application of the Navier-Stokes and continuity equations. The convergence criterion was set based on previous models and convergence studies. The residuals used to compute the flow equations were defined with an absolute criteria of 10⁻³, and the number of iterations per time-step was set to 200, with a time-step size of 0.01 s.

Cats were all modelled with a heart rate of 140 beats per minute (bpm) over the course of three cardiac cycles. The first cardiac cycle was considered as a warm-up, while the second one was used for analysis. The boundary conditions (shown in Fig. 6) were defined in all analysed cases as a generic pressure inlet at the PVs and a velocity outlet at the MV, which is the scenario providing more realistic simulations in sensitivity analysis performed in human data¹⁵. The MV velocity profile was obtained from Schober and Chetboul¹⁰ using a Doppler echocardiographic image of a feline heart with cardiomyopathy (Fig. 6b). The data was extracted using WebPlotDigitizer (<https://apps.automeris.io/wpd4/>). According to Fig. 6b, the normal maximum velocity was 1 m/s, and the E-wave to A-wave ratio was within the normal range of 1 to 2. The transmitral velocity profile (Fig. 6c) was consistent across all feline hearts analyzed, although the flow through the mitral valve varied depending on the valve area. Its profile was applied like top-hat, where kinetic energy is uniformly distributed across the mitral valve cross-section. This approach is widely employed in CFD to simulate valve outflow conditions in a controlled and repeatable manner. Additionally, a generic pressure waveform, also reported in Schober and Chetboul¹⁰, was applied as the PV inlet boundary condition for all studied cases (Fig. 6a).

We used a spring diffusion-based dynamic mesh approach available in Ansys Fluent to impose movement to the MV annulus ring plane, following measurements from Spalla et al.⁴⁰. This strategy, mimicking the passive movement of the MV ring induced by the contraction of the left ventricle, provides more realistic blood flow patterns than with rigid walls, as demonstrated in human LA fluid simulations¹⁴. Figure S1 in the Supplementary Material shows an example of left atrial wall motion along the cardiac cycle obtained with the dynamic mesh approach. The convergence criterion was set based on previous models and convergence studies, ensuring consistency with our established CFD pipeline. The simulations were performed in serial on a local workstation equipped with an Intel(R) Core(TM) i7-9700 CPU @ 3.00 GHz processor, 32 GB of RAM, and running Windows

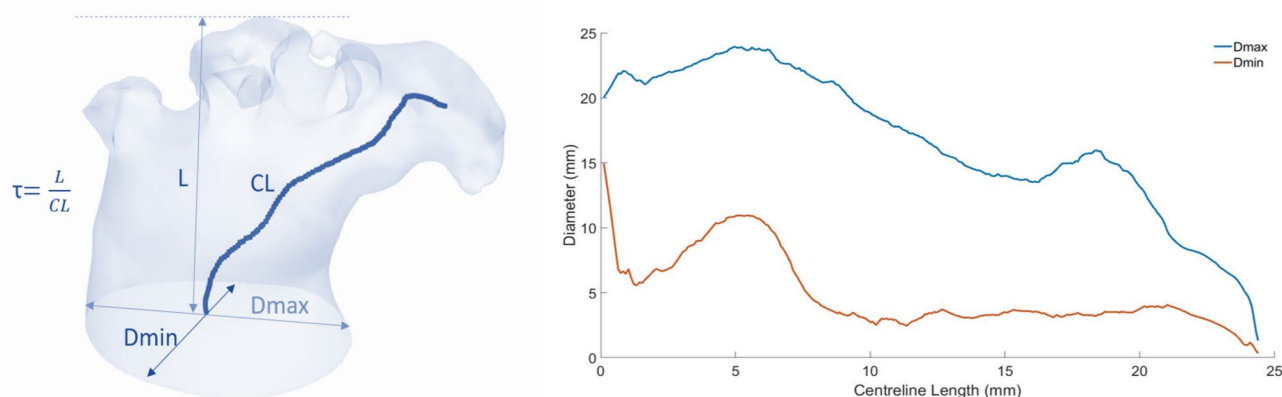


Fig. 5. Left atrial appendage (LAA) geometry with morphological parameters indicated, including the centreline (CL) and tortuosity (τ). The graph on the right illustrates the progression of local diameters, with maximum and minimum values shown in blue and orange respectively along the centreline length, displaying a continuous reduction towards the endpoint of the LAA.

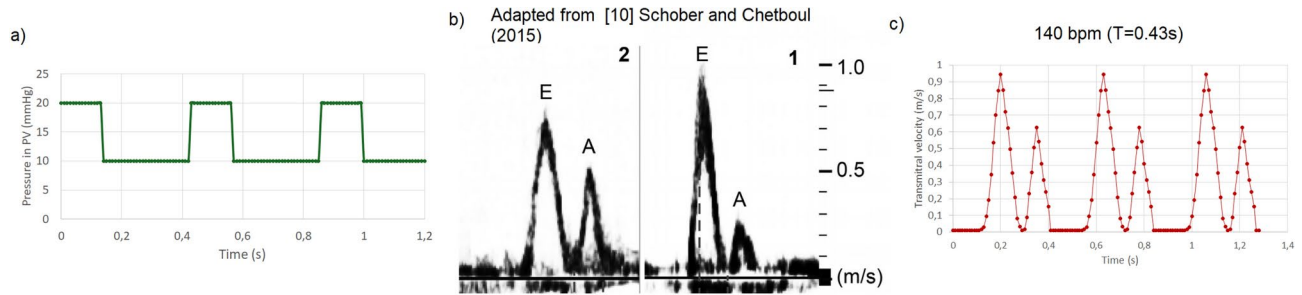


Fig. 6. Boundary conditions used at the inlets and outlets of feline hearts for fluid simulations. **(a)** Generic pressure waveform applied as the inlet (pulmonary veins) boundary condition. **(b)** Doppler velocity profile from echocardiography at the mitral valve (MV) of a cardiomyopathy cat adapted from Schober and Chetboul¹⁰. **(c)** Transmitral flow velocity is applied as the generic outflow condition for all left atrial assessments in cats.

10 Pro as the operating system. Depending on the complexity of the cases, the simulation times ranged from approximately 4 to 9 hours.

Haemodynamic indices

Blood flow velocities

Blood flow velocities were derived from the CFD simulations, focusing on the LAA ostium since it has traditionally been used in human data to identify cases with a higher risk of thrombus formation. As per Virchow's triad, one of the primary contributors to thrombus formation is blood stasis near the LA surface. Consequently, low blood flow velocities in the LAA ostium at end-diastole (0.2 m/s as in Tamura et al.⁴¹) have been associated with an increased thrombotic risk. In our study, we integrate the blood flow velocities from the fluid simulations across the whole ostium at end-diastole, and we normalize them with the LA volume to consider variations due to LA remodelling and heart sizes, as in Pons et al.¹².

Flow metrics at the left atrial wall

From the wall shear stress (WSS) data, several parameters related to thrombus formation were calculated, including the oscillatory shear index (OSI) and the endothelial cell activation potential (ECAP). It is well established that cases with low blood velocities and complex flow patterns are more susceptible to thrombus formation¹³. All these parameters were computed using Matlab R2018b (R2018b Academic license).

The time-averaged wall shear stress magnitude (TAWSS) was defined as:

$$TAWSS = \frac{1}{T} \int_0^T |WSS| dt, \quad (1)$$

where T represents the duration of the cardiac cycle. TAWSS characterizes the mechanobiological effects of WSS on the LAA wall. Lower values of WSS correspond to reduced flow velocities, which are associated with blood stasis and a higher risk of coagulation.

The OSI is a non-dimensional parameter that characterizes the predominant direction of blood flow during the cardiac cycle and is defined as:

$$OSI = \frac{\frac{1}{2} \int_0^T WSS dt}{\int_0^T |WSS| dt}. \quad (2)$$

In essence, OSI estimates how much the direction of the WSS field changes due to complex flow patterns and has a maximum value of 0.5.

The ECAP is calculated as the ratio between OSI and TAWSS:

$$ECAP = \frac{OSI}{TAWSS}, \quad (3)$$

and it can help identify regions within the LA that are at a higher risk of thrombus formation. High ECAP values are associated with high OSI (complex flows) and low TAWSS (low blood flow velocities). To filter out extremely low values related to the high trabeculations detected in the LAA, a threshold of TAWSS < 0.01 Pa was applied during the analysis.

Shear rate

Virchow's triad includes the analysis of potential changes in blood viscosity⁴². Several fundamental factors contribute to these changes, including plasma viscosity, haematocrit, cell deformability and aggregability, as discussed in previous studies^{43,44}.

However, it is important to note that the present study lacks specific data on the blood properties of each cat. Most simulations assumed an asymptotic viscosity for blood (approximately 0.0035 Pa s). This assumption is based on the premise that shear rates are high enough to neglect shear thinning, allowing the application of the Navier-Stokes equations. Nonetheless, it is worth considering that shear rates can influence blood viscosity. In humans, shear rates below 100 s^{-1} are associated with non-Newtonian fluid behaviour and increased viscosity⁴⁵. The rheological behaviour of blood in cats as in other species, is linked to the shear field generated under high shear rates⁴⁶.

Statistical analysis

The statistical analysis was conducted using R Studio 1.2.1335 (<https://rstudio.com>). Student t-test, Mann-Whitney-Wilcoxon test and Chi-squared test were applied as appropriate. The normality of the data was assessed using the Shapiro-Wilk test. The level of significance (α) was set at 0.05, and all statistical tests were two-sided.

Several comparisons were made to identify shape differences between the different groups, with the following number of cases in each group: normal (5) vs. cardiomyopathy (16); thrombus (6) vs. non-thrombus (10); asymptomatic HCM (5) vs. symptomatic (CHF-Thrombus; 11).

For the analysis of the haemodynamic parameters (ECAP and OSI), a sampling approach was employed. All mean values were normalized using a minimum-maximum approach. The point closest to the 90th percentile value within the distribution was selected. This point represents the maximum value within the distribution, allowing the capture of potentially significant patterns in each group. Additionally, the average velocity at the ostium during the second cardiac cycle was analysed.

Data availability

Data supporting these findings are available from the corresponding author upon request.

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Author contributions

Conceptualisation, J.N.M., A.L.O., M.I.P., J.M., O.C., V.L.F.; methodology, J.M., M.I.P. and A.L.O.; software, J.M., A.L.O., X.M. and M.I.P.; formal analysis, M.I.P., J.M. and A.L.O.; investigation, J.M., A.L.O. and V.A.; data curation, clinical support and data acquisition, J.N.M., I.C., A.G. and V.L.F.; image processing, P.G.C., A.C.C., I.C.S., O.J.A.; writing original draft preparation, M.I.P. and A.L.O.; writing-review and editing, M.I.P., A.L.O., J.M., O.C.; visualisation, A.L.O., X.M. and M.I.P. All authors have read and agreed to the published version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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